

Name of Student:		Class: BTech	
Enrollment No:		Batch	
Date of Experiment	Date of Submission	Submitted on:	
Remarks by faculty:		Grade:	
Signature of student:		Signature of Faculty:	

1. Case study of evolution of Operating System w.r.t Computer Generations.

Sr. No.	Computer Generation	Significant Aspects of Computer Generation	OS Generation	Significant Aspects of OS	Year
1	1st Generation	Vacuum Tubes, Machine Language, Punched Cards, Batch Processing	No Formal OS	Simple routines, Machine language programming, Direct hardware control, No abstraction	1940s-1950s
2	2nd Generation	Transistors, Assembly Language, Magnetic Tape/Disk, Batch Processing	Early Batch Systems	Resident Monitors, Simple batch processing, Job sequencing, Limited resource management	1950s-1960s

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3	3rd Generation	Integrated Circuits, High-Level Languages, Multiprogramming, Time-Sharing	Multiprogramming/Time-Sharing OS	Multiprogramming, Time-sharing, Spooling, Resource allocation, Complex scheduling, Early forms of memory management	1960s-1970s
4	4th Generation	Microprocessors, Personal Computers, Networking, GUIs	Personal/Network OS	User-friendly interfaces (GUIs), Network support, Distributed systems, Microcomputer OS, Increased portability, File systems	1970s-1980s
5	5th Generation	Parallel Processing, AI, Mobile Computing,	Distributed/Mobile/Cloud OS	Advanced resource management, Virtualization, Cloud	1980s-Present

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		Cloud Computing, IoT		computing integration, Mobile OS, AI integration, Security advancements, Scalability	
6	6th Generation(Future)	Quantum Computing, Advanced AI, Neuromorphic Computing, Hyperconnectivity	Emerging OS	Quantum algorithms, Advanced AI-driven OS, Hyperconnected networks, Neuromorphic interfaces	TBD